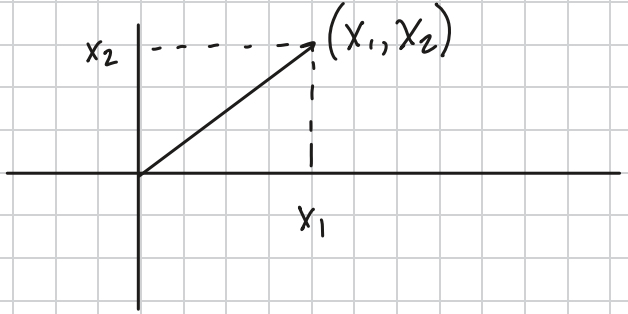
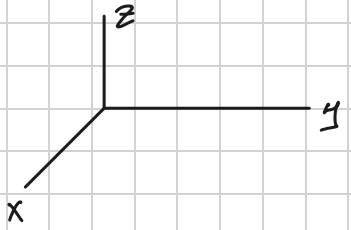
Week 7

$\mathbb{R}^2 \rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$





Standard bases

$\mathbb{R}^2$ Standard basis vectors

$$\begin{bmatrix}1\\0\end{bmatrix} \qquad \begin{bmatrix}0\\1\end{bmatrix}$$

e_1 e_2

Standard

Standard

$$\begin{bmatrix}x_1\\x_2\end{bmatrix} = x_1 \cdot \begin{bmatrix}1\\0\end{bmatrix} + x_2 \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}x_1\\0\end{bmatrix} + \begin{bmatrix}0\\x_2\end{bmatrix} = \begin{bmatrix}x_1\\x_2\end{bmatrix}$$

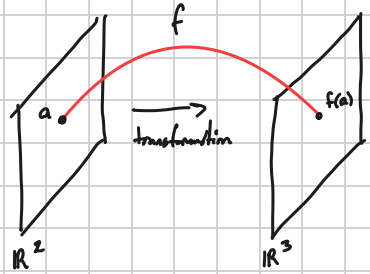
decompose or write in terms of linear comb of basis vectors

$$\begin{bmatrix}3\\1\end{bmatrix} = 3 \cdot e_1 + 1 \cdot e_2$$

Coefficients

$n=1$ $[x]$

+ define transformations/functions



$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \begin{matrix} -? \\ -? \\ -? \end{matrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Example $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} x_1 + 2x_2 \\ -x_2 \\ x_1^2 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} 1 + 2 \cdot 0 \\ -0 \\ 1^2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

"how many transformations are there?"

matrix tranformations

T_A ← matrix that does the transfer

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \xrightarrow{T_A} \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = Y$$

$\mathbb{R}^n$ $\mathbb{R}^m$

$\dim(A) = m \times n$

$$\overline{X} \xrightarrow{T_A} \overline{Y} = A \cdot \overline{X}$$

matrix transformation

$\mathbb{R}^n \xrightarrow{T_A} \mathbb{R}^m$ is a matrix multiplication

Checks

$\dim(Y) = m \times 1$ (output)

$\dim(X) = n \times 1$ (input)

$$\overline{Y} = A \cdot \overline{X}$$

$$(m \times n) \cdot (n \cdot 1) = (m \times 1)$$

$\dim(A)$ $\dim(X)$

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 0 \\ 4 & 3 \end{bmatrix}$$

$\dim(A) = 3 \times 2$

$\quad\quad\quad

Matrix transformations are linear (will be on midterm)

rules to be satisfied

① $T_A(k \cdot \vec{x}) = k \cdot T_A(\vec{x})$

② $T_A(X + Y) = T_A(X) + T_A(Y)$

Ex

$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \xrightarrow{T_A} \begin{bmatrix} \pi \\ -\frac{1}{2} \\ 7 \end{bmatrix}$

if you know its is linear

② $\cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix} \rightarrow 2 \begin{bmatrix} \pi \\ -\frac{1}{e} \\ 7 \end{bmatrix} = \begin{bmatrix} 2\pi \\ -\frac{2}{e} \\ 14 \end{bmatrix}$

$\begin{bmatrix} 2 \\ 6 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \xrightarrow{T_A} \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$

$\begin{bmatrix} -1 \\ 1 \end{bmatrix} \xrightarrow{T_A} \begin{bmatrix} 0 \\ 6 \\ 7 \end{bmatrix}$

$\begin{bmatrix} 0 \\ 2 \end{bmatrix} \xrightarrow{?}$

$\begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad 2T \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ Die

Any matrix transformation is linear

Examples for midterm

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \longrightarrow \begin{bmatrix} x_1 + 2x_2 \\ 0 \end{bmatrix}$$

$\mathbb{R}^2$ $\qquad$ $\mathbb{R}^2$

Proof (it is linear)

- $\begin{bmatrix} k \cdot x_1 \\ k \cdot x_2 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} kx_1 + 2(kx_2) \\ 0 \end{bmatrix} = k \cdot \begin{bmatrix} x_1 + 2x_2 \\ 0 \end{bmatrix}$ (factor out)

- $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \xrightarrow{T}$

$\downarrow$

$$\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} (x_1 + y_1) + 2(x_2 + y_2) \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 + 2x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} y_1 + 2y_2 \\ 0 \end{bmatrix}$$

$$T\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad T\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$